ROBERT AFIAKWA OWUSU

Biomedical Engineer | Computational Bioengineering | AI in Healthcare

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PROFESSIONAL SUMMARY

Biomedical engineer interested in understanding complex biological systems through computational approaches. My research combines artificial intelligence, molecular modeling, and biomedical data analysis to address challenges in therapeutic discovery and healthcare.

EDUCATION

Kwame Nkrumah University of Science and Technology, Ghana **Jan 2022 – Nov 2025**

B.Sc. Biomedical Engineering, First Class Honours

- GPA: 3.98/4.00; Valedictorian, Department of Computer Engineering.

RESEARCH INTERESTS

Computational Biology; Computational Modeling; Medical Imaging; Biomedical Signal Processing; Optics and Photonics.

TECHNICAL SKILLS

Research and Data Analysis: Literature review, scientific writing, study design, data curation and standardization, statistical analysis, data visualization, and research presentation.

Computational Methods: Machine learning, deep learning, biomedical image analysis, molecular docking, virtual screening, protein–ligand interaction analysis, and physiological signal processing.

Programming and Engineering Tools: Python, pandas, NumPy, scikit-learn, PyRx, AutoDock Vina, PyMOL, UCSF Chimera, BIOVIA Discovery Studio, 3D Slicer, and SOLIDWORKS.

RESEARCH EXPERIENCE

Research Assistant | Global Health and Infectious Diseases (GHID) AI Team, KCCR **Oct 2025 – Present**

Supervisors: Prof. Edmund Ekuadzi, Prof. John H. Amuasi, and Dr. Prince Ebenezer Adjei

- **AI-Powered Natural Product Systems Pharmacology:** Building the West African Traditional Medicine System Pharmacology (WATMSP) platform by integrating phytochemical, molecular, target, disease, ADMET, ethnomedicinal, and therapeutic indication data for West African medicinal plants (*manuscript in preparation*).
- **Cardiotoxicity Prediction and Molecular Docking:** Applying machine-learning and LLM-assisted workflows to evaluate bioactive molecules across hERG, Nav1.5, and Cav1.2 endpoints, together with molecular docking and protein–ligand interaction analysis.
- **ACE-Targeted Antihypertensive Drug Discovery:** Curating and standardizing ACE bioactivity records from ChEMBL and PubChem for machine-learning, ligand-based modeling, and lead prioritization.

Research Trainee | Spark Academy Imaging Research Program **Jan 2026 – Jun 2026**

- Co-authored a workshop paper evaluating breast ultrasound lesion segmentation under progressive speckle corruption, submitted to the **MICCAI 2026 MIRASOL Workshop**.
- Benchmarked U-Net, Attention U-Net, TransUNet, and SegFormer-B0 and applied Monte Carlo Dropout for predictive uncertainty and model reliability assessment.

Undergraduate Thesis | Low-Cost Portable Wireless Digital Stethoscope **Jan 2025 – Aug 2025**

Supervisor: Dr. Prince Ebenezer Adjei

- Developed a portable digital stethoscope integrating biomedical signal acquisition and wireless transmission for remote auscultation in resource-limited clinical environments.
- Collaborated with clinicians to evaluate device usability and refine the prototype for cardiovascular monitoring and telemedicine.

Research Assistant | Machine Learning Model Applications in Medicine **Sep 2024 – Oct 2024**

Supervisor: Dr. Daniel Akwei Addo

- Contributed to review work on machine-learning model applications in cardiovascular medicine and on artificial intelligence models for predicting outcomes of lung pathologies.
- Conducted literature searches, synthesized evidence across biomedical studies, and assessed model performance, interpretability, generalizability, and dataset limitations.

SELECTED COMPUTATIONAL RESEARCH PROJECTS

Structure-Based Virtual Screening of NDM-1 Inhibitors **Mar 2026 – Apr 2026**

- Developed and validated a PyRx/AutoDock Vina workflow to screen candidate NDM-1 inhibitors and characterize active-site, metal-coordination, and protein–ligand interactions using PyMOL and BIOVIA Discovery Studio.

Trustworthy Deep Learning for Brain Tumor Segmentation Using Multi-Modal MRI **Apr 2026 – Jun 2026**

- Developed transfer-learning pipelines using the BraTS dataset to segment brain tumors from multi-modal MRI and evaluated model performance, robustness, and predictive uncertainty using clinically relevant segmentation metrics.

Molecular Modeling and Dynamics of KRAS G12D Inhibitors **Jul 2026 – Present**

- Comparing candidate and reference inhibitors through molecular docking and residue-level interaction profiling, while preparing selected protein–ligand complexes for molecular dynamics simulation and trajectory-based stability analysis.

PUBLICATIONS AND MANUSCRIPTS

- "Beyond Dice: Evaluating Reliability and Robustness in Breast Ultrasound Lesion Segmentation Under Progressive Speckle Corruption." Submitted to the **MICCAI 2026 MIRASOL Workshop**.
- "West African Traditional Medicine Systems Pharmacology Platform." *Manuscript in preparation.*

TEACHING AND PROFESSIONAL EXPERIENCE

Teaching Assistant | Department of Computer Engineering, KNUST **Oct 2025 – Present**

- Selected as Teaching Assistant based on outstanding academic performance to support Biosignals Analysis, Digital Signal Processing, and Biomedical Optics through tutorials, laboratory instruction, instructional material development, and undergraduate capstone mentoring.

Biomedical Engineering Intern | Komfo Anokye Teaching Hospital **Sep 2023 – Nov 2023**

- Performed medical-equipment inspections, preventive maintenance, troubleshooting, inventory documentation, and clinical-user support.

Biomedical Engineering Intern | Swedru Municipal Hospital **Oct 2022 – Dec 2022**

- Assisted with medical-equipment installation, routine inspection, maintenance, troubleshooting, and inventory management.

SELECTED CONFERENCES & PRESENTATIONS

- **Noguchi Annual Research Meeting**
  Poster Presenter, Accra, Ghana — Nov 2025
- **Frontiers in Optics + Laser Science Conference** (Optica)
  Participant, Denver, Colorado, USA — Oct 2025
- **Technology Week, KNUST**
  Poster Presenter — *A Low-Cost Portable Wireless Digital Stethoscope for Clinical Use in Ghana* — Aug 2025

SELECTED HONORS AND AWARDS

- **Valedictorian, Department of Computer Engineering, KNUST** — Nov 2025
- **Best Biomedical Engineering Student, College of Engineering Provost Awards** — July 2023 – 2026
- **Optica Amplify Immersion Program Scholar, Denver, Colorado, USA.** — Oct 2025
- **Winner, Ghana National Gas Challenge** — Nov 2024
- **Ghana Education Trust Fund Scholar** — April 2022
- **Best graduating Student, Sweru Senior High School** — Nov 2020
- **Winner, Second Ghana Science Olympiad** — Dec 2019

LEADERSHIP AND PROFESSIONAL MEMBERSHIPS

Academic Head, Ghana Engineering Students Association; Clinic Project Team Lead, Engineers Without Borders–KNUST; former Vice President, Biomedical Engineering Students Society. Member of Optica, SPIE, NSBE, Black in AI.